注意：

允許學生個人、非營利性的圖書館或公立學校合理使用本基金會網站所提供之各項試題及其解答。**可直接下載而不須申請**。

重版、系統地複製或大量重製這些資料的任何部分，必須獲得財團法人臺北市九章數學教育基金會的授權許可。

申請此項授權請電郵 ccmp@seed.net.tw

Notice:

Individual students, nonprofit libraries, or schools are permitted to make fair use of the papers and its solutions. Republication, systematic copying, or multiple reproduction of any part of this material is permitted only under license from the Chiuchang Mathematics Foundation.

Requests for such permission should be made by e-mailing Mr. Wen-Hsien SUN ccmp@seed.net.tw

2012 IMSO Answers

SHORT ANSWER

1.	300%	**2.**	1155	**3.**	8	**4.**	15	**5.**	22
6.	5	**7.**	49	**8.**	4	**9.**	9	**10.**	12
11.	145	**12.**	63 or 99	**13.**	15689	**14.**	65°	**15.**	7
16.	91	**17.**	343	**18.**	8	**19.**	42	**20.**	7
21.	7 : 18	**22.**	79800	**23.**	4486	**24.**	A	**25.**	15

ESSAY PROBLEMS

1.	270	**2.**	55 km per hour	**3.**	67	**4.**	24
5.	24	**6.**	7.5	**7.**	C	**8.**	$\frac{6}{13}$

9.	4	**10.**	60°	**11.**	55	**12.**	1512	**13.**	3, 6 and 10

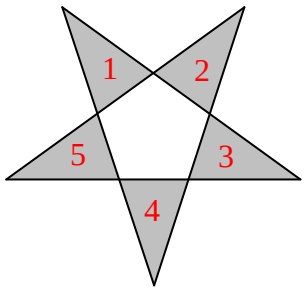
EXPLORATION PROBLEMS

1. Integers: $\frac{4}{1}, \frac{30}{6}, \frac{28}{14}, \frac{27}{9}, \frac{26}{13}, \frac{25}{5}, \frac{24}{12}, \frac{22}{11}, \frac{21}{7}, \frac{20}{10}, \frac{18}{2}, \frac{16}{8}, \frac{15}{3}$

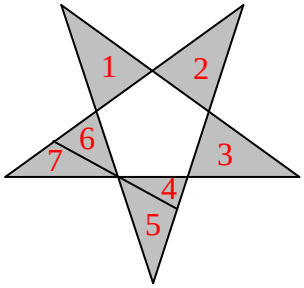
Non-integers: $\frac{29}{23}, \frac{19}{17}$

2. 21912, 23232, 25652, 29392, 40304, 42724, 44044, 46464, 48884, 61116, 63536, 65956, 67276, 69696, 80608, 84348, 86768, 88088.

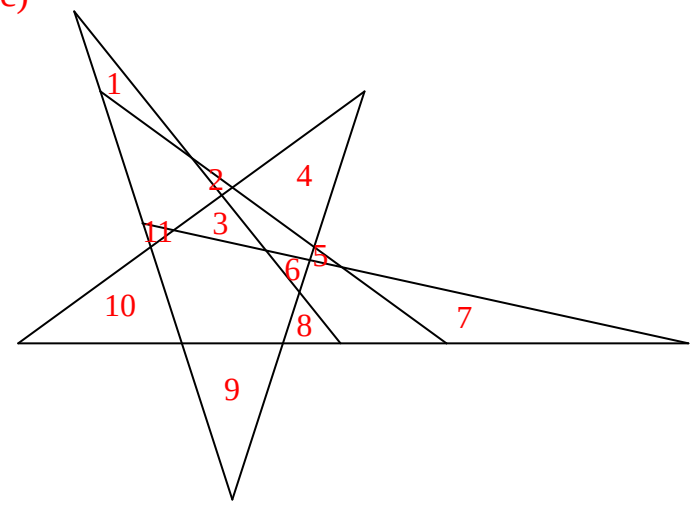
3. (a)



(b)



(c)



4. 7

5. 199

6.

10	11	20	21	22	23	42
9	12	19	18	17	24	41
8	13	14	15	16	25	40
7	30	29	28	27	26	39
6	31	32	33	34	35	38
5	4	3	2	1	36	37